

Technology Stack

Date	5 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>Streamlit</i>	<i>Streamlit allows for rapid development of data-driven web applications using pure Python. It automatically handles responsive UI design, interactive widgets, and session state without needing to write complex HTML/CSS/JS.</i>

2	Backend / Server-Side	<i>Python</i>	<i>Python is the industry standard for Artificial Intelligence. It provides seamless integration with our custom BiLSTM/BERT emotion detection models and APIs, ensuring robust server-side processing.</i>
3	Database / Data Storage	<i>SQLite & CSV (Pandas)</i>	<i>SQLite provides a lightweight, serverless relational database perfect for securely storing user credentials and interaction history. CSV files are used for flexible data handling and data analysis.</i>
4	Cloud / Hosting / Deployment	<i>Local Environment</i> <i>(with potential for Streamlit Community Cloud)</i>	<i>Running locally allows for secure, cost-effective development and rapid testing of heavy machine learning models without incurring expensive cloud server costs.</i>

5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Git ensures safe version tracking of our code and model changes, while GitHub provides remote repository backup and a professional platform to showcase the project portfolio.</i>
6	Third-Party APIs / Other Tools	<i>Google Gemini API, TensorFlow/Keras, HuggingFace, Plotly</i>	• Gemini API: Generates empathetic learning support responses. • TensorFlow & HuggingFace: Powers the core BiLSTM and BERT emotion detection models. • Plotly: Renders interactive charts for the user's Analytics Dashboard